

SECTION 4

NORMAL PROCEDURES

4.1 GENERAL

This section contains instructions and recommended procedures that are peculiar to the operation of this helicopter.

For definition of terms, abbreviations and symbols used in this section refer to Section 1.

4.2 PREPARATION FOR FLIGHT

4.2.1 Flight Restrictions

The minimum, normal, maximum and cautionary operation ranges for the helicopter and its subsystems are indicated by instrument markings, placards and decals.

For helicopter and subsystem restrictions refer to Section 2, Limitations.

4.2.2 Flight Planning

Refer to Section 5 and 9 to determine required fuel, airspeeds and power settings for takeoff, climb, cruise, and hovering and landing data necessary to accomplish the mission.

4.2.3 Mass and Balance

The takeoff and anticipated landing gross mass and balance should be obtained before takeoff and checked against mass and load limits, and center of gravity limitations (see Section 2).

4.3 PREFLIGHT CHECK

4.3.1 General

The preflight check shall be accomplished, at the latest, before the first flight of the day and according to either the Flight Manual, the Maintenance Manual or the Pilot's Checklist.

The preflight check is not a detailed mechanical inspection, but essentially a visual check of the helicopter for correct condition.

Items marked with a star (★) need to be checked each flight before takeoff.

When unusual local conditions dictate, the extent and/or frequency of this check shall be increased as necessary to promote safe operation.

NOTE • The following list contains only check items for the clean configuration.

- In addition to these items, check antennas and all installed optional equipment. Make certain that relevant intermediate and special inspections, in accordance with the Maintenance Manual, are complied with.
- For optional equipment check items refer to the respective Flight Manual Supplement or Maintenance Manual Chapter 800.

4.3.2 Exterior Check

The exterior check is laid out as a walk-around check, starting forward right at the pilot's door, proceeding clockwise to the tail boom, to the left hand side (including the upper and lower areas of the helicopter) and being completed at the helicopter nose area.

NOTE • The helicopter should be headed into the wind before starting engines.

- The area around the helicopter should be clear of all foreign objects.
- To avoid excessive drain on the helicopter battery, particularly during cold weather, all ground operations should be conducted using an external power unit (EPU).
- When the battery is used, the operation of electrical equipment should be kept to a minimum.

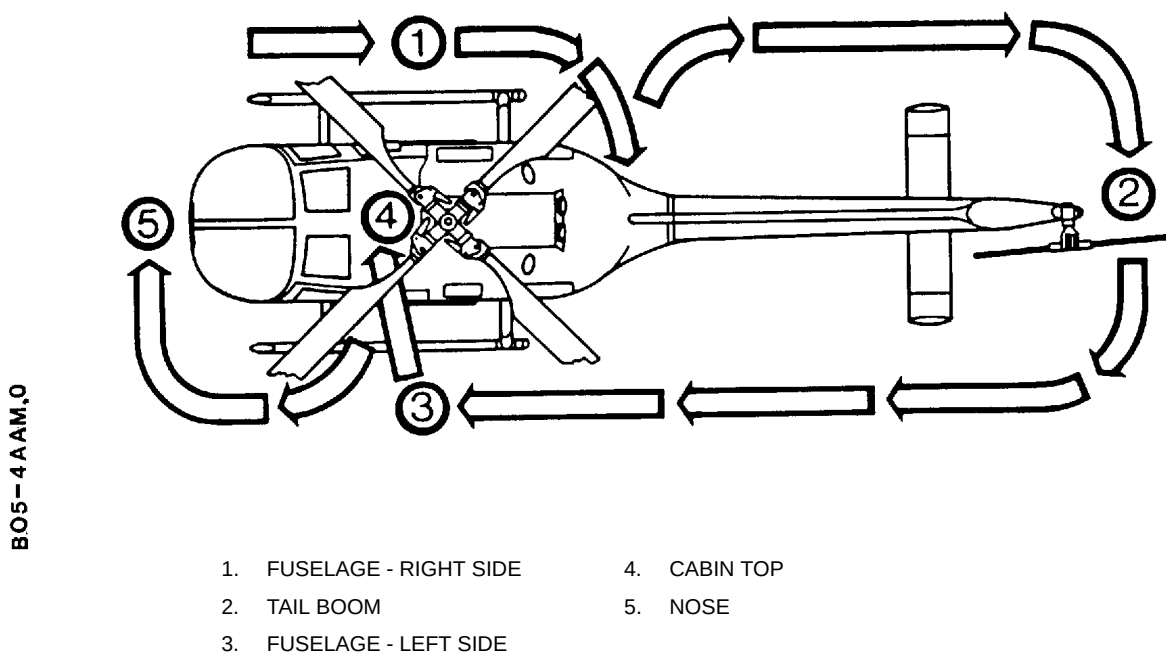


Fig. 4-1 Exterior Check Sequence

- ★ Helicopter forms and documents
 - Check, complete
- Fuselage underside
 - Condition, no fuel leaks
- Each fuel tank (3)
 - Drain sample
- When OAT is above 40 °C or below –32 °C:
 - Fuel temperature
 - Measure; determine fuel type to be used and/or the permissible altitude in accordance with Section 2.
- ★ Covers and tie-downs
 - Removed
- ★ Ice and snow (if any)
 - Removed
- ★ Ground handling wheels
 - Removed

Cockpit

- All switches
 - Off
- Power levers
 - OFF
- Rotor brake lever (if installed)
 - Full down
- Circuit breakers
 - As required
- Battery switch
 - On
- Voltmeter indication
 - 24V DC nominal
- If EPU is used:
 - Battery switch
 - Off
 - Pitot tube heating
 - Function

Lighting:

- Position lights
 - Function
- Anti-collision lights
 - Function
- Landing light(s)
 - Function

Before Night Flights:

- Dome and instrument lights
 - Function
- Utility lamp
 - Function
- Hand lamp
 - Function
- Battery switch
 - Off
- Hand fire extinguisher
 - On board, check pressure
- Flashlight
 - On board, function

First aid kit (if installed here)

– On board

Fuselage - Right Side

Cockpit overhead window

– Condition

★ OAT indicator probe

– Condition

Cockpit door

– Condition, function, jettison handle secured

Pilot seat and safety belts

– Condition

★ Static port

– Clean, unobstructed

★ Pitot tube(s)

– Condition, clean, unobstructed

Landing gear and step

– Condition

Cabin door

– Condition, function

★ Baggage, cargo, loose items

– Stowed, secured

Passenger seats and seat belts

– Condition

★ Transmission compartment

– No foreign objects, no leakage, free of ice/snow

EFFECTIVITY *For helicopters after SB BO105–10–126*

Transmission oil filter
clogging indicator pin

– Check in

EFFECTIVITY *ALL*

Oil cooler

– No leakage

★ Engine oil levels

– Above ADD OIL mark

Engine oil tanks

– Security of mounting, no leakage, filler caps secured

Scavenge oil filter (if installed) clog-
ging indicator pin

– Check in

★ Transmission oil level

– Above MIN mark

Transmission

– Condition, security of mounting, no leakage, filler cap closed

Rotor brake (if installed)

– Condition

★ Engine air intake

– Condition, no dirt, no foreign objects, free of ice/snow

Compressor

– Rotate by hand, check for smooth rotation

★ Transmission access door

– Closed, secured

Fuselage - Right Side (Continued)

★ Engine compartment	– No foreign objects, no fuel/oil leakage
★ Engine compartment ventilation air intake	– Clear
Engine	– Condition
Wiring, linkages and lines	– Condition, no leakage, no chafing
Engine mounts	– Condition, security of mounting
N ₁ -governor and connections	– Security of mounting
Starter/Generator	– Security of mounting
Engine driveshaft	– Check freewheeling
★ Engine access door	– Closed, secured
Drain and vent ports	– Clear
Battery (if installed here)	– Condition, no electrolyte spillage, security of mounting
Battery connectors (2)	– Connected
Battery vents	– Clear
★ Battery compartment door	– Closed, secured
Engine cowling	– Secured
Engine exhaust pipe	– Condition, security of mounting
★ Fuselage exterior	– Condition
Clam shell doors and latches	– Condition, function
First aid kit (if installed here)	– On board
Avionics (in avionics equipment bay)	– Security of mounting
★ Baggage, cargo, loose items	– Stowed, secured
Avionics bay door	– Closed
★ Clam shell doors	– Closed, secured
Spoiler	– Condition
Tail Boom	
★ Tail boom - RH	– Condition
★ Tail-rotor driveshaft fairing	– Secured
Drain ports	– Clear

Tail Boom (Continued)

- | | |
|---|------------------------------------|
| ★ RH horizontal and vertical stabilizer, position light (green) | — Condition |
| Vertical fin, position light (white), anti-collision light | — Condition |
| Vertical fin cowling | — Secured |
| ★ Tail rotor | — Condition |
| ★ Pitch links | — Condition |
| PU Erosion-protective film (if fitted) | — Condition, no separation |
| ★ Intermediate gearbox | — Oil level, no leakage |
| ★ Tail-rotor gearbox | — Condition, oil level, no leakage |

NOTE Oil levels may be easier to verify by grasping tail skid and shortly shaking tail boom.

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|---|-------------------|
| ★ Intermediate gearbox access door (if installed) | — Closed, secured |
| Tail skid | — Condition |
| ★ LH horizontal and vertical stabilizer, position light (red) | — Condition |
| ★ Tail boom - LH | — Condition |
| ★ Tail-rotor driveshaft fairing | — Secured |
| Drain ports | — Clear |

Fuselage - Left Side

- | | |
|---|---|
| ★ Fuel filler cap | — Closed, secured |
| Drain and vent ports | — Clear |
| ★ Fuselage exterior | — Condition |
| Engine exhaust pipe | — Condition, security of mounting |
| Engine cowling | — Secured |
| ★ Engine compartment | — No foreign objects, no fuel/oil leakage |
| ★ Engine compartment ventilation air intake | — Clear |
| Engine | — Condition |
| Wiring, linkages and lines | — Condition, no leakage, no chafing |
| Engine mounts | — Condition, security of mounting |
| N ₂ -governor and actuator | — Condition, security of mounting |

Fuselage - Left Side (Continued)

Starter/Generator	– Security of mounting
Engine driveshaft	– Check freewheeling
★ Engine access door	– Closed, secured
★ Hydraulic compartment	– No foreign objects, no leakage, free of ice/snow
★ Hydraulic module	– No leakage
★ Area between hydraulic module and XMSN floor, especially control rod bellows	– Free of ice/snow
Water drain ports	– Clear
Hydraulic filter clogging indicator pins (4)	– Check in
★ Hydraulic reservoir oil levels	– Above FILL mark
★ Service-fitting caps	– Installed
Transmission	– Condition, security of mounting, no leakage
★ Engine air intake	– Condition, no dirt, no foreign objects, free of ice/snow
Compressor	– Rotate by hand, check for smooth rotation
★ Hydraulic compartment access door	– Closed, secured
Main rotor	– Rotate by hand in direction of rotation, check for smooth rotation
★ Rotor hub oil level	– Between MIN and MAX marks
★ Rotor blades	– Condition
★ Rotor blade attachment bolts	– Secured
PU Erosion-protective film (if fitted)	– Condition, no separation
★ Driving link assy, mixing lever assy	– Condition, secured
★ Rotating control rods	– Condition, free movement, secured
Rotating control rod spherical bearings	– Smooth operation
Swashplate and boot (if installed)	– Condition, secured
Pendulum absorbers (if installed)	– Condition, free movement
★ Air intake	– Condition, no foreign objects, free of ice/snow

Fuselage - Left Side (Continued)

Engine inlet deflector shield (if installed)	— Condition, no foreign objects
Cockpit overhead window	— Condition
Cabin door	— Condition, function
★ Baggage, cargo, loose items	— Stowed, secured
Passenger seats and seat belts	— Condition
Fuel vent port	— Clear
Landing gear and step	— Condition
★ Static port	— Clean, unobstructed
Cockpit door	— Condition, function, jettison handle secured
Copilot cyclic stick (if installed)	— Secured, safety-wired
Copilot collective lever (if installed)	— Secured
Copilot seat and safety belts	— Condition
★ If copilot seat is unoccupied: Copilot safety belts	— Fastened, tightened

Nose

Windshields	— Condition
Battery (if installed here)	— Condition, no electrolyte spillage, security of mounting
Battery connectors (2)	— Connected
Battery vents	— Clear
Nose compartment	— No foreign objects
★ Nose compartment door	— Closed, secured
Cockpit air intake screens	— Clear
Drain ports	— Clear
Lower cockpit windows	— Condition
Fuselage bottom	— Condition
Anti-collision light	— Condition
Landing light(s)	— Condition

4.3.3 Interior Check

- ★ Seat and pedals – Adjust
- ★ Safety belts – Fasten, adjust
- ★ Shoulder harness locks – Function
- ★ Instrument panel:
 - All instruments – Check and set
- ★ Switch panel:
 - Emergency fuel shutoff switches – Guarded
 - Circuit breakers – As required
 - All switches – Off
 - COMM/NAV equipment – Off
- ★ Overhead panel:
 - Power levers – OFF
 - All switches – Off
 - Circuit breakers – As required
- ★ Collective lever panel:
 - All switches – Off

Pre-start Check

- ★ Battery switch – On
 - For EPU starts:
 - Battery switch – Off
 - EPU – Connected
 - EPU ON** caution light – On
- ★ Voltmeter indication :
 - Battery – Minimum 24V
 - EPU – Between 24-28V
- ★ Warning/caution lights – Test
- ★ Mast moment indicator – Test
- ★ Fuel quantity indicator – Check readings
- ★ Fuel pumps (4) – Function
- ★ Collective lever lock – Release
- ★ Flight controls – Check for freedom of movement, full travel

Hydraulic override (HY BLOCK)

 – Check function

NOTE To achieve correct results, cyclic stick must be moved exactly along the longitudinal and lateral axes of the helicopter.

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|---|---|
| Collective lever | – Raise slightly |
| Hydraulic test switch | – RESET and release |
| HY BLOCK warning light | – Check off |
| Collective lever | – Raise until HY BLOCK warning light comes on, then lower slightly and release |
| Hydraulic test switch | – RESET and release |
| HY BLOCK warning light | – Check off |
| Collective lever | – Lower until HY BLOCK warning light comes on, then raise slightly and release |
| Hydraulic test switch | – RESET and release |
| HY BLOCK warning light | – Check off |
| Cyclic stick | – Move forward until HY BLOCK warning light comes on and release |
| Hydraulic test switch | – RESET and release |
| HY BLOCK warning light | – Check off |
| Each remaining cyclic control direction | – Check as above |
| Collective lever | – Full down |
| ★ Collective lever lock | – Engage |
| ★ HY BLOCK warning light | – Check on |
| <u>Cyclic trim system</u> | – <u>Check function</u> |

NOTE This check ensures that cyclic stick full travel is still guaranteed after the trim actuators have reached their full travel stop.

- | | |
|---|--|
| Cyclic trim switch | – Press forward and hold until actuator audibly stops |
| Cyclic stick | – Move fore and aft to limit stops; check for full fore/aft travel and free movement |
| Each remaining cyclic control direction | – Check as above |
| ★ Cyclic stick | – Centered |
| ★ Rotor brake lever (if fitted) | – Full down |